INFORMATION SHEET

90nm COMPLETE DESIGN KIT (CDK090)

The Cadence 90nm generic Complete Design Kit (CDK) includes a standard cell library, IO cell library, and a process design kit (PDK). The CDK contains proprietary Cadence intellectual property, but does not include any third party proprietary information.

The 90nm CDK is being developed to support Virtuoso Flow Solutions Ether and Raptor projects. The 90nm CDK and these projects will be used to develop flows, demos and workshops as follows:

Virtuoso Demos/Workshops	CDK090 Support
Virtuoso AMS Block Authoring	Yes
Virtuoso AMS Top-Down Sim.	Yes
Virtuoso AMS Chip Finishing	Yes
Virtuoso RFIC Flow	No

This information sheet (Version 3.2) describes the contents and status at the beginning of Q405. A list of frequently asked questions is included at the end of this document.

Table 1 Standard Cell and IO Views

Description	Extension	Std Cell gsclib090	I/O Cells giolib090	Comments
Netlist Data				
- Verilog Macro	*.v	Yes	Yes	Each cell has a Verilog macro definition
- Spice Netlist	*.spi	Yes	No	Each cell has a Spice netlist
- Extracted Netlist	*.sp	Yes	Yes	Netlists were extracted from layouts
CDB Library Data				
- DFII Technology File	*.tf	N/A	N/A	Included in GPDK090 process design kit library
- DFII Schematic	schematic view	Yes	Yes	Custom Schematics
- DFII Symbol	symbol view	Yes	Yes	Custom or auto-generated Symbols
- DFII Layout	layout view	Yes	Yes	Each cell has custom layout
- DFII Abstract	abstract view	Yes	Yes	
OA 2.0 Library Data				
- DFII Technology File	*.tf	N/A	N/A	Included in GPDK090 process design kit library
- Schematic	schematic view	Yes	Yes	
- Symbol	symbol view	Yes	Yes	
- Layout	layout view	Yes	Yes	
- DFII Abstract	abstract view	Yes	Yes	
OA 2.2 Library Data				
- DFII Technology File	*.tf	N/A	N/A	Included in GPDK090 process design kit library
- Schematic	schematic view	Yes	Yes	
- Symbol	symbol view	Yes	Yes	
- Layout	layout view	Yes	Yes	
- DFII Abstract	abstract view	Yes	Yes	
Archival & Exchange Formats				
- EDIF Symbols	*.edf	No	No	EDIF is not used in the Cadence flows to date
- GDS	*.gds	Yes	Yes	
- LEF abstracts	*.lef	Yes	Yes	Technology LEF supplied with gsclib090
- Antenna LEF	*_ant.lef	Yes	No	IO ant.lef not needed or planned for Ether/Raptor projects

Table 1 (continued) Standard Cell and IO Views

Description	Extension	Std Cell gsclib090	I/O Cells giolib090	Comments
Timing Data				
- Liberty format	*.lib	Yes	Yes	Refer to Table 2
- TLF format	*.tlf	Yes	Yes	Refer to Table 2
Other Cadence Tool Data				
- Signal Storm Library	*.alf	No	No	Experimental, not used in current Ether/Raptor
- Celtic	*.cdB	Yes	Yes	Experimental, not used in current Ether/Raptor

Timing data availability for the Standard Cells and IOs is listed in Table 2. It's important to note that the Ether/Raptor projects do not have any timing requirements and as a result the data has not been tested or validated.

Table 2 Standard Cell and IO Characterization

Characterization Table	Liberty	TLF	Comments
Standard Cells			
- Slow/Slow 125C 0.9V	Yes	Yes	Experimental, not used in current Ether/Raptor projects
- Fast/Fast 0C 1.1V	Yes	Yes	Experimental, not used in current Ether/Raptor projects
- Typ/Typ 25C 1.0V	Yes	Yes	Experimental, not used in current Ether/Raptor projects
I/O Cells			
- Slow/Slow 125C 0.9V & 2.25V	Yes	Yes	Experimental, not used in current Ether/Raptor projects
- Fast/Fast 0C 1.1V & 2.75V	Yes	Yes	Experimental, not used in current Ether/Raptor projects
- Typ/Typ 25C 1.0V & 2.5V	Yes	Yes	Experimental, not used in current Ether/Raptor projects

A list of the standard cells and IOs are provided separately in Tables 4 and 6.

Table 3 Process Design Kit (PDK)

Description	Extension	Files in PDK	Comments
Technology files			
- DFII Technology file (binary, ASCII)	*.tf, *.cds	Yes	Techfile objects needed for RF flow are not planned
- DFII Display.drf file (ASCII)	*.drf	Yes	
- LEF Technology file (ASCII)	*.lef	Yes	**Tech. LEF from gsclib090 was LEFIN'd to gpdk090**
DFII CDB, OA 2.0 & OA2.2 Pcells			
- mosfet (pmos, nmos)(hvt, 1v, 2v)	(in tf)	Yes	
- bjt (vpnp, pnp, npn)	(in tf)	Yes	
- diode	(in tf)	Yes	
- moscap	(in tf)	Yes	
- poly resistor	(in tf)	Yes	
- symbolic (vias, contacts)	(in tf)	Yes	
- mimcap	(in tf)	No	Not needed or planned for Ether/Raptor projects
- rf mosfets (pmos, nmos)	(in tf)	No	Not needed or planned for Ether/Raptor projects
- varactor	(in tf)	No	Not needed or planned for Ether/Raptor projects
- nwell resistor	(in tf)	No	Not needed or planned for Ether/Raptor projects
- inductor	(in tf)	No	Not needed or planned for Ether/Raptor projects
Simulation Models			
- mosfet (pmos, nmos)(hvt, 1v, 2v)	*.scs	Yes	Refer to Table 3a, Simulation
- bjt (vpnp, npn, pnp)	*.scs	Yes	Refer to Table 3a, Simulation

Table 3 (continued) Process Design Kit (PDK)

Description	Extension	Files in PDK	Comments
- diode	*.scs	Yes	Refer to Table 3a, Simulation
- moscap	*.scs	Yes	Refer to Table 3a, Simulation
- poly resistor	*.scs	Yes	Refer to Table 3a, Simulation
- mimcap	*.scs	No	Not needed or planned for Ether/Raptor projects
- rf mosfets (pmos, nmos)	*.scs	No	Not needed or planned for Ether/Raptor projects
- varactor	*.scs	No	Not needed or planned for Ether/Raptor projects
- nwell resistor	*.scs	No	Not needed or planned for Ether/Raptor projects
- inductor	*.scs	No	Not needed or planned for Ether/Raptor projects
Physical Verification Rule Decks			
- Diva DRC	*.rul	Yes	Antenna, density, drc and drc.rsf
- Diva LVS	*.rul	Yes	Extraction, lvs
- Assura DRC	*.rul	Yes	Antenna, density, drc
- Assura LVS	*.rul	Yes	Extraction,Compare, LVS and LVSinclude.rsf, bind.rul
- Assura RCX	*.rcx	Yes	proc, cap2d, capgen command rule
- Assura HF	?	No	Not needed or planned for Ether/Raptor projects
- Assura RF	*.rul	No	Not needed or planned for Ether/Raptor projects
- Calibre DRC/LVS/RCX	?	No	Not needed or planned for Ether/Raptor projects
- PVS DRC	drc.rul	Yes	Not tested for Ether/Raptor projects
- Hercules DRC/LVS/RCX	?	No	Not needed or planned for Ether/Raptor projects
Archival & Exchange Files			
- GDSII	streamLayer.Map	Yes	One layer-mapping file for both streamIn & streamOut
Other Cadence Tool Data			
- NeoCircuit	NeoDevice	Yes	
- NeoCell	NeoConfig	Yes	
- Virtuoso Layout Migrate	*.qtt, migratemap*.il	Yes	
- Virtuoso Chip Assembly Router	icc.rules	Yes	
- VoltageStorm (VSPE)	*.cl & .tch	No	Will be needed for Q405 Ether project; VSPE requires power information in the *.lib files
- VoltageStorm (VAVO)	*emData	Yes	Experimental, planned for Q405 Ether project.
- Fire & Ice QX	*.ict, *.layermap	Yes	Experimental, not used in current Ether/Raptor flows
- SNA	*.cds, *.enc	Yes	Experimental, not used in current Ether/Raptor flows
- SOCE	*.capTbl	Yes	

Table 3a Simulation

Simulation Models	BSIM3	Corners	Statistical (Mismatch)	Statistical (Process Variation)
- mosfet (pmos, nmos)(hvt, 1v, 2v)	Yes	FF/SS/TT/FS/SF	No	No
- rf mosfets (pmos, nmos)	No	FF/SS/TT/FS/SF	No	No
- bjt (vpnp)	Yes	FF/SS/TT/FS/SF	No	No
- diode	Yes	FF/SS/TT/FS/SF	No	No
- mimcap	Yes	FF/SS/TT/FS/SF	No	No
- moscap	Yes	FF/SS/TT/FS/SF	No	No
- varactor	Yes	FF/SS/TT/FS/SF	No	No
- poly resistor	Yes	FF/SS/TT/FS/SF	No	No
- nwell resistor	Yes	FF/SS/TT/FS/SF	No	No
- inductor	Yes	FF/SS/TT/FS/SF	No	No

The generic 90nm IO power requirements are 1.0 volt for core power and 2.5 volts for ESD & IO drive. The drive strength is fixed at 2ma and is non-programmable. A list of the available IO cells is provided in Table 4.

Table 4 IO Cell Features

Cell Name	Description	non-Stagger bond	Stagger bond	HVT	non-HVT	ESD Spec (KV & V)
I/O FEED CELLS						
- IORINGFEED0005	Feed cell, width 0.005 um (1 manufacturing unit wide)	Yes	No	No	Yes	N/A
- IORINGFEED001	Feed cell, width 0.01 um wide	Yes	No	No	Yes	N/A
- IORINGFEED01	Feed cell, width 0.1 um wide	Yes	No	No	Yes	N/A
- IORINGFEED1	Feed cell, width 1 um wide	Yes	No	No	Yes	N/A
- IORINGFEED3	Feed cell, width 3 um wide	Yes	No	No	Yes	N/A
- IORINGFEED3x	Isolate PVDD/PVSS domain from other PVDD/PVSS domains, width 3um wide	Yes	No	No	Yes	N/A
- IORINGFEED5	Feed cell, width 5 um wide	Yes	No	No	Yes	N/A
- IORINGFEED10	Feed cell, width 10 um wide	Yes	No	No	Yes	N/A
- IORINGFEED60	Feed cell, width 60 um wide	Yes	No	No	Yes	N/A
- IORINGCORNER	Corner cell	Yes	No	No	Yes	N/A
I/O+PAD POWER CELLS						
- PADVDD	Core and IO power (1.0 volt)	Yes	No	No	Yes	Untested
- PADVDD25	Core power (1.0 to 2.5 volt)	Yes	No	No	Yes	Untested
- PADVSS	Core and IO ground	Yes	No	No	Yes	Untested
- PADVSS25	Core ground					
- PADVDDIOR	2.5 volt IO Ring, ESD and pad power	Yes	No	No	Yes	Untested
- PADVSSIOR	IO Ring, ESD and pad ground	Yes	No	No	Yes	Untested
I/O+PAD DIGITAL CELLS						
- PADDB	PAD, Bidirectional	Yes	No	No	Yes	Untested
- PADDI	PAD, Input	Yes	No	No	Yes	Untested
- PADDO	Pad, Output (output 2ma)	Yes	No	No	Yes	Untested
- PADDOZ	Pad, Output, (output 2ma) Tri-state high impedance	Yes	No	No	Yes	Untested
- PADDIS	Pad, Schmitt-Trigger Input	No	No	No	No	Untested
- PADDBS	Pad, Bidirectional (input/output 2ma) Schmitt-Trigger Input	No	No	No	No	Untested
I/O+PAD ANALOG CELLS						
- PADANALOG	Pad, Analog (range 0 to 2.5 volts)	Yes	No	No	Yes	Untested
I/O+PAD RF CELLS						
	Not required for Ether and Raptor projects	No	No	No	No	Untested
I/O's W/O PAD CELLS						
- IORING*	For each PAD* (I/O+PAD) cell above, the corresponding I/O cell name has the prefix "IORING" instead of "PAD"	Same	No	No	Same	Untested

Table 5 Standard Cells

Cell Family	Cell Type	Drive Strength	Prefix List
And	52	1,2,4,6,8,L	AND(18), NAND(34)
Or	64	1,2,4,6,8,L	OR(18), NOR(34), XOR(6), XNOR(6)
Inverter	10	1,2,3,4,6,8,12,16,20,L	INV(10)
Buffer/Tri-State-Buffer w/ & w/o enable/inverter	18	1,2,3,4,6,8,12,16,20,L	BUF(8), TBUF(10)
Clock Buffer/Inverted Clock Buffer/Gated Clock Latch	33	1,2,3,4,6,8,12,16,20	CLKBUF(8),CLKAND(6), CLKINV(9), CLKXOR(4), CLKMX(6)
AOI/OAI/AO/OA	96	1,2,4,L	AOI(40), OAI(40), AO(8), OA(8)
Half/Full Adder	12	1,2,4,L	ADDF(8), ADDH(4)
Mux with/without inverted output	28	1,2,4,6,8,L	MX(28)
DFF: pos-edge, neg-edge, async/sync R/S, w/o R/S	44	1,2,4,8,L	DFF(44)
Enable DFF; async/sync R, w/o R	16	1,2,4,8,L	EDFF(12), MDFF(4)
Scan version of all DFFs	60	1,2,4,8,L	SDFF(44), SEDFF(12), SMDFF(4)
Latch: active high/low enable, async R/S	32	1,2,3,4,6,8,12,16,20,L	TLAT(32)
Balanced Rise/Fall INV/NAND/AND/MUX/XOR	2	2,4	BMXI(2)
Dlay/Tie-High/Tie-Low	10	1,4, HI, LO	DLY(8), TIE(2)
Filler Cells	7	1,2,4,8,16,32,64	FILL(7)
Antenna cell/Decoupling cells	1	1	ANT(1)
Other cells	4	1	HOLD(1), VDD(1), VSS(1), ACH(1)
Total Cell Number	*489*		30 basic types

Table 6 List of Standard Cells

List of Standard Cells	Description
AND2X1/	AND logic gate
AND2X2/	AND logic gate
AND2X4/	AND logic gate
AND2X6/	AND logic gate
AND2X8/	AND logic gate
AND2XL/	AND logic gate
AND3X1/	AND logic gate
AND3X2/	AND logic gate
AND3X4/	AND logic gate
AND3X6/	AND logic gate
AND3X8/	AND logic gate
AND3XL/	AND logic gate
AND4X1/	AND logic gate
AND4X2/	AND logic gate
AND4X4/	AND logic gate
AND4X6/	AND logic gate
AND4X8/	AND logic gate
AND4XL/	AND logic gate
18	
NAND2BX1/	NOT AND logic gate
NAND2BX2/	NOT AND logic gate
NAND2BX4/	NOT AND logic gate
NAND2BXL/	NOT AND logic gate
NAND2X1/	NOT AND logic gate
NAND2X2/	NOT AND logic gate
NAND2X4/	NOT AND logic gate
NAND2X6/	NOT AND logic gate
NAND2X8/	NOT AND logic gate
NAND2XL/	NOT AND logic gate
NAND3BX1/	NOT AND logic gate
NAND3BX2/	NOT AND logic gate
NAND3BX4/	NOT AND logic gate
NAND3BXL/	NOT AND logic gate
NAND3X1/	NOT AND logic gate
NAND3X2/	NOT AND logic gate
NAND3X4/	NOT AND logic gate
NAND3X6/	NOT AND logic gate
NAND3X8/	NOT AND logic gate
NAND3XL/	NOT AND logic gate
NAND4BBX1/	NOT AND logic gate
NAND4BBX2/	NOT AND logic gate
NAND4BBX4/	NOT AND logic gate
NAND4BBXL/	NOT AND logic gate
NAND4BX1/	NOT AND logic gate
NAND4BX2/	NOT AND logic gate
NAND4BX4/	NOT AND logic gate
NAND4BXL/	NOT AND logic gate
NAND4X1/	NOT AND logic gate
NAND4X2/	NOT AND logic gate
NAND4X4/	NOT AND logic gate
NAND4X6/	NOT AND logic gate
NAND4X8/	NOT AND logic gate
NAND4XL/	NOT AND logic gate
34	

List of Standard Cells	Description
OR2X1/	OR logic gate
OR2X2/	OR logic gate
OR2X4/	OR logic gate
OR2X6/	OR logic gate
OR2X8/	OR logic gate
OR2XL/	OR logic gate
OR3X1/	OR logic gate
OR3X2/	OR logic gate
OR3X4/	OR logic gate
OR3X6/	OR logic gate
OR3X8/	OR logic gate
OR3XL/	OR logic gate
OR4X1/	OR logic gate
OR4X2/	OR logic gate
OR4X4/	OR logic gate
OR4X6/	OR logic gate
OR4X8/	OR logic gate
OR4XL/	OR logic gate
18	
NOR2BX1/	NOT OR logic gate
NOR2BX2/	NOT OR logic gate
NOR2BX4/	NOT OR logic gate
NOR2BXL/	NOT OR logic gate
NOR2X1/	NOT OR logic gate
NOR2X2/	NOT OR logic gate
NOR2X4/	NOT OR logic gate
NOR2X6/	NOT OR logic gate
NOR2X8/	NOT OR logic gate
NOR2XL/	NOT OR logic gate
NOR3BX1/	NOT OR logic gate
NOR3BX2/	NOT OR logic gate
NOR3BX4/	NOT OR logic gate
NOR3BXL/	NOT OR logic gate
NOR3X1/	NOT OR logic gate
NOR3X2/	NOT OR logic gate
NOR3X4/	NOT OR logic gate
NOR3X6/	NOT OR logic gate
NOR3X8/	NOT OR logic gate
NOR3XL/	NOT OR logic gate
NOR4BBX1/	NOT OR logic gate
NOR4BBX2/	NOT OR logic gate
NOR4BBX4/	NOT OR logic gate
NOR4BBXL/	NOT OR logic gate
NOR4BX1/	NOT OR logic gate
NOR4BX2/	NOT OR logic gate
NOR4BX4/	NOT OR logic gate
NOR4BXL/	NOT OR logic gate
NOR4X1/	NOT OR logic gate
NOR4X2/	NOT OR logic gate
NOR4X4/	NOT OR logic gate
NOR4X6/	NOT OR logic gate
NOR4X8/	NOT OR logic gate
NOR4XL/	NOT OR logic gate
34	

Table 6 (continued) List of Standard Cells

List of Standard Cells	Description
XOR2X1/	EXCLUSIVE OR logic gate
XOR2X2/	EXCLUSIVE OR logic gate
XOR2X4/	EXCLUSIVE OR logic gate
XOR2XL/	EXCLUSIVE OR logic gate
XOR3X1/	EXCLUSIVE OR logic gate
XOR3XL/	EXCLUSIVE OR logic gate
6	
XNOR2X1/	EXCLUSIVE NOT OR logic gate
XNOR2X2/	EXCLUSIVE NOT OR logic gate
XNOR2X4/	EXCLUSIVE NOT OR logic gate
XNOR2XL/	EXCLUSIVE NOT OR logic gate
XNOR3X1/	EXCLUSIVE NOT OR logic gate
XNOR3XL/	EXCLUSIVE NOT OR logic gate
6	
INVX1	INVERTER
INVX12	INVERTER
INVX16	INVERTER
INVX2	INVERTER
INVX20	INVERTER
INVX3	INVERTER
INVX4	INVERTER
INVX6	INVERTER
INVX8	INVERTER
INVXL	INVERTER
10	
BUFX12/	BUFFER
BUFX16/	BUFFER
BUFX2/	BUFFER
BUFX20/	BUFFER
BUFX3/	BUFFER
BUFX4/	BUFFER
BUFX6/	BUFFER
BUFX8/	BUFFER
8	
TBUFX1/	TRI-STATE BUFFER
TBUFX12/	TRI-STATE BUFFER
TBUFX16/	TRI-STATE BUFFER
TBUFX2/	TRI-STATE BUFFER
TBUFX20/	TRI-STATE BUFFER
TBUFX3/	TRI-STATE BUFFER
TBUFX4/	TRI-STATE BUFFER
TBUFX6/	TRI-STATE BUFFER
TBUFX8/	TRI-STATE BUFFER
TBUFXL/	TRI-STATE BUFFER
10	
CLKBUFX12/	CLOCKED BUFFER
CLKBUFX16/	CLOCKED BUFFER
CLKBUFX2/	CLOCKED BUFFER
CLKBUFX20/	CLOCKED BUFFER
CLKBUFX3/	CLOCKED BUFFER
CLKBUFX4/	CLOCKED BUFFER
CLKBUFX6/	CLOCKED BUFFER
CLKBUFX8/	CLOCKED BUFFER
8	

List of Standard Cells	Description
CLKAND2X12/	CLOCKED AND
CLKAND2X2/	CLOCKED AND
CLKAND2X3/	CLOCKED AND
CLKAND2X4/	CLOCKED AND
CLKAND2X6/	CLOCKED AND
CLKAND2X8/	CLOCKED AND
6	
CLKINVX1/	CLOCKED INVERTER
CLKINVX12/	CLOCKED INVERTER
CLKINVX16/	CLOCKED INVERTER
CLKINVX2/	CLOCKED INVERTER
CLKINVX20/	CLOCKED INVERTER
CLKINVX3/	CLOCKED INVERTER
CLKINVX4/	CLOCKED INVERTER
CLKINVX6/	CLOCKED INVERTER
CLKINVX8/	CLOCKED INVERTER
9	
CLKXOR2X1/	CLOCKED EXCLUSIVE OR
CLKXOR2X2/	CLOCKED EXCLUSIVE OR
CLKXOR2X4/	CLOCKED EXCLUSIVE OR
CLKXOR2X8/	CLOCKED EXCLUSIVE OR
4	
CLKMX2X12/	CLOCKED MULTIPLEXOR
CLKMX2X2/	CLOCKED MULTIPLEXOR
CLKMX2X3/	CLOCKED MULTIPLEXOR
CLKMX2X4/	CLOCKED MULTIPLEXOR
CLKMX2X6/	CLOCKED MULTIPLEXOR
CLKMX2X8/	CLOCKED MULTIPLEXOR
6	
OA21X1/	OR-AND logic
OA21X2/	OR-AND logic
OA21X4/	OR-AND logic
OA21XL/	OR-AND logic
OA22X1/	OR-AND logic
OA22X2/	OR-AND logic
OA22X4/	OR-AND logic
OA22XL/	OR-AND logic
8	
AO21X1/	AND-OR logic
AO21X2/	AND-OR logic
AO21X4/	AND-OR logic
AO21XL/	AND-OR logic
AO22X1/	AND-OR logic
AO22X2/	AND-OR logic
AO22X4/	AND-OR logic
AO22XL/	AND-OR logic
8	

Table 6 (continued) List of Standard Cells

List of Standard Cells	Description
AOI211X1/	AND-OR-INVERT logic
AOI211X2/	AND-OR-INVERT logic
AOI211X4/	AND-OR-INVERT logic
AOI211XL/	AND-OR-INVERT logic
AOI21X1/	AND-OR-INVERT logic
AOI21X2/	AND-OR-INVERT logic
AOI21X4/	AND-OR-INVERT logic
AOI21XL/	AND-OR-INVERT logic
AOI221X1/	AND-OR-INVERT logic
AOI221X2/	AND-OR-INVERT logic
AOI221X4/	AND-OR-INVERT logic
AOI221XL/	AND-OR-INVERT logic
AOI222X1/	AND-OR-INVERT logic
AOI222X2/	AND-OR-INVERT logic
AOI222X4/	AND-OR-INVERT logic
AOI222XL/	AND-OR-INVERT logic
AOI22X1/	AND-OR-INVERT logic
AOI22X2/	AND-OR-INVERT logic
AOI22X4/	AND-OR-INVERT logic
AOI22XL/	AND-OR-INVERT logic
AOI2BB1X1/	AND-OR-INVERT logic
AOI2BB1X2/	AND-OR-INVERT logic
AOI2BB1X4/	AND-OR-INVERT logic
AOI2BB1XL/	AND-OR-INVERT logic
AOI2BB2X1/	AND-OR-INVERT logic
AOI2BB2X2/	AND-OR-INVERT logic
AOI2BB2X4/	AND-OR-INVERT logic
AOI2BB2XL/	AND-OR-INVERT logic
AOI31X1/	AND-OR-INVERT logic
AOI31X2/	AND-OR-INVERT logic
AOI31X4/	AND-OR-INVERT logic
AOI31XL/	AND-OR-INVERT logic
AOI32X1/	AND-OR-INVERT logic
AOI32X2/	AND-OR-INVERT logic
AOI32X4/	AND-OR-INVERT logic
AOI32XL/	AND-OR-INVERT logic
AOI33X1/	AND-OR-INVERT logic
AOI33X2/	AND-OR-INVERT logic
AOI33X4/	AND-OR-INVERT logic
AOI33XL/	AND-OR-INVERT logic
40	

List of Standard Cells	Description
OAI211X1/	OR-AND-INVERT logic
OAI211X2/	OR-AND-INVERT logic
OAI211X4/	OR-AND-INVERT logic
OAI211XL/	OR-AND-INVERT logic
OAI21X1/	OR-AND-INVERT logic
OAI21X2/	OR-AND-INVERT logic
OAI21X4/	OR-AND-INVERT logic
OAI21XL/	OR-AND-INVERT logic
OAI221X1/	OR-AND-INVERT logic
OAI221X2/	OR-AND-INVERT logic
OAI221X4/	OR-AND-INVERT logic
OAI221XL/	OR-AND-INVERT logic
OAI222X1/	OR-AND-INVERT logic
OAI222X2/	OR-AND-INVERT logic
OAI222X4/	OR-AND-INVERT logic
OAI222XL/	OR-AND-INVERT logic
OAI22X1/	OR-AND-INVERT logic
OAI22X2/	OR-AND-INVERT logic
OAI22X4/	OR-AND-INVERT logic
OAI22XL/	OR-AND-INVERT logic
OAI2BB1X1/	OR-AND-INVERT logic
OAI2BB1X2/	OR-AND-INVERT logic
OAI2BB1X4/	OR-AND-INVERT logic
OAI2BB1XL/	OR-AND-INVERT logic
OAI2BB2X1/	OR-AND-INVERT logic
OAI2BB2X2/	OR-AND-INVERT logic
OAI2BB2X4/	OR-AND-INVERT logic
OAI2BB2XL/	OR-AND-INVERT logic
OAI31X1/	OR-AND-INVERT logic
OAI31X2/	OR-AND-INVERT logic
OAI31X4/	OR-AND-INVERT logic
OAI31XL/	OR-AND-INVERT logic
OAI32X1/	OR-AND-INVERT logic
OAI32X2/	OR-AND-INVERT logic
OAI32X4/	OR-AND-INVERT logic
OAI32XL/	OR-AND-INVERT logic
OAI33X1/	OR-AND-INVERT logic
OAI33X2/	OR-AND-INVERT logic
OAI33X4/	OR-AND-INVERT logic
OAI33XL/	OR-AND-INVERT logic
40	

Table 6 (continued) List of Standard Cells

List of Standard Cells	Description
ADDFHX1/	FULL ADDER
ADDFHX2/	FULL ADDER
ADDFHX4/	FULL ADDER
ADDFHXL/	FULL ADDER
ADDFX1/	FULL ADDER
ADDFX2/	FULL ADDER
ADDFX4/	FULL ADDER
ADDFXL/	FULL ADDER
8	
ADDHX1/	HALF ADDER
ADDHX2/	HALF ADDER
ADDHX4/	HALF ADDER
ADDHXL/	HALF ADDER
4	
MX2X1/	MULTIPLEXOR
MX2X2/	MULTIPLEXOR
MX2X4/	MULTIPLEXOR
MX2X6/	MULTIPLEXOR
MX2X8/	MULTIPLEXOR
MX2XL/	MULTIPLEXOR
MX3X1/	MULTIPLEXOR
MX3X2/	MULTIPLEXOR
MX3X4/	MULTIPLEXOR
MX3XL/	MULTIPLEXOR
MX4X1/	MULTIPLEXOR
MX4X2/	MULTIPLEXOR
MX4X4/	MULTIPLEXOR
MX4XL/	MULTIPLEXOR
MXI2X1/	MULTIPLEXOR
MXI2X2/	MULTIPLEXOR
MXI2X4/	MULTIPLEXOR
MXI2X6/	MULTIPLEXOR
MXI2X8/	MULTIPLEXOR
MXI2XL/	MULTIPLEXOR
MXI3X1/	MULTIPLEXOR
MXI3X2/	MULTIPLEXOR
MXI3X4/	MULTIPLEXOR
MXI3XL/	MULTIPLEXOR
MXI4X1/	MULTIPLEXOR
MXI4X2/	MULTIPLEXOR
MXI4X4/	MULTIPLEXOR
MXI4XL/	MULTIPLEXOR
28	
BMXIX2/	MULTIPLEXOR
BMXIX4/	MULTIPLEXOR
2	

List of Standard Cells	Description
DFFHQX1/	D-FLIP-FLOP
DFFHQX2/	D-FLIP-FLOP
DFFHQX4/	D-FLIP-FLOP
DFFHQX8/	D-FLIP-FLOP
DFFNSRX1/	D-FLIP-FLOP
DFFNSRX2/	D-FLIP-FLOP
DFFNSRX4/	D-FLIP-FLOP
DFFNSRXL/	D-FLIP-FLOP
DFFQX1/	D-FLIP-FLOP
DFFQX2/	D-FLIP-FLOP
DFFQX4/	D-FLIP-FLOP
DFFQXL/	D-FLIP-FLOP
DFFRHQX1/	D-FLIP-FLOP
DFFRHQX2/	D-FLIP-FLOP
DFFRHQX4/	D-FLIP-FLOP
DFFRHQX8/	D-FLIP-FLOP
DFFRX1/	D-FLIP-FLOP
DFFRX2/	D-FLIP-FLOP
DFFRX4/	D-FLIP-FLOP
DFFRXL/	D-FLIP-FLOP
DFFSHQX1/	D-FLIP-FLOP
DFFSHQX2/	D-FLIP-FLOP
DFFSHQX4/	D-FLIP-FLOP
DFFSHQX8/	D-FLIP-FLOP
DFFSRHQX1/	D-FLIP-FLOP
DFFSRHQX2/	D-FLIP-FLOP
DFFSRHQX4/	D-FLIP-FLOP
DFFSRHQX8/	D-FLIP-FLOP
DFFSRX1/	D-FLIP-FLOP
DFFSRX2/	D-FLIP-FLOP
DFFSRX4/	D-FLIP-FLOP
DFFSRXL/	D-FLIP-FLOP
DFFSX1/	D-FLIP-FLOP
DFFSX2/	D-FLIP-FLOP
DFFSX4/	D-FLIP-FLOP
DFFSXL/	D-FLIP-FLOP
DFFTRX1/	D-FLIP-FLOP
DFFTRX2/	D-FLIP-FLOP
DFFTRX4/	D-FLIP-FLOP
DFFTRXL/	D-FLIP-FLOP
DFFX1/	D-FLIP-FLOP
DFFX2/	D-FLIP-FLOP
DFFX4/	D-FLIP-FLOP
DFFXL/	D-FLIP-FLOP
44	

Table 6 (continued) List of Standard Cells

List of Standard Cells	Description
EDFFHQX1/	ENABLE D-FLIP-FLOP
EDFFHQX2/	ENABLE D-FLIP-FLOP
EDFFHQX4/	ENABLE D-FLIP-FLOP
EDFFHQX8/	ENABLE D-FLIP-FLOP
EDFFTRX1/	ENABLE D-FLIP-FLOP
EDFFTRX2/	ENABLE D-FLIP-FLOP
EDFFTRX4/	ENABLE D-FLIP-FLOP
EDFFTRXL/	ENABLE D-FLIP-FLOP
EDFFX1/	ENABLE D-FLIP-FLOP
EDFFX2/	ENABLE D-FLIP-FLOP
EDFFX4/	ENABLE D-FLIP-FLOP
EDFFXL/	ENABLE D-FLIP-FLOP
12	
MDFFHQX1/	MUX D-FLIP-FLOP
MDFFHQX2/	MUX D-FLIP-FLOP
MDFFHQX4/	MUX D-FLIP-FLOP
MDFFHQX8/	
4	
SEDFFHQX1/	SCAN ENABLE D-FLIP-FLOP
SEDFFHQX2/	SCAN ENABLE D-FLIP-FLOP
SEDFFHQX4/	SCAN ENABLE D-FLIP-FLOP
SEDFFHQX8/	SCAN ENABLE D-FLIP-FLOP
SEDFFTRX1/	SCAN ENABLE D-FLIP-FLOP
SEDFFTRX2/	SCAN ENABLE D-FLIP-FLOP
SEDFFTRX4/	SCAN ENABLE D-FLIP-FLOP
SEDFFTRXL/	SCAN ENABLE D-FLIP-FLOP
SEDFFX1/	SCAN ENABLE D-FLIP-FLOP
SEDFFX2/	SCAN ENABLE D-FLIP-FLOP
SEDFFX4/	SCAN ENABLE D-FLIP-FLOP
SEDFFXL/	SCAN ENABLE D-FLIP-FLOP
12	
SMDFFHQX1/	SCAN MUX D-FLIP-FLOP
SMDFFHQX2/	SCAN MUX D-FLIP-FLOP
SMDFFHQX4/	SCAN MUX D-FLIP-FLOP
SMDFFHQX8/	SCAN MUX D-FLIP-FLOP
4	

List of Standard Cells	Description
SDFFHQX1/	SCAN D-FLIP-FLOP
SDFFHQX2/	SCAN D-FLIP-FLOP
SDFFHQX4/	SCAN D-FLIP-FLOP
SDFFHQX8/	SCAN D-FLIP-FLOP
SDFFNSRX1/	SCAN D-FLIP-FLOP
SDFFNSRX2/	SCAN D-FLIP-FLOP
SDFFNSRX4/	SCAN D-FLIP-FLOP
SDFFNSRXL/	SCAN D-FLIP-FLOP
SDFFQX1/	SCAN D-FLIP-FLOP
SDFFQX2/	SCAN D-FLIP-FLOP
SDFFQX4/	SCAN D-FLIP-FLOP
SDFFQXL/	SCAN D-FLIP-FLOP
SDFFRHQX1/	SCAN D-FLIP-FLOP
SDFFRHQX2/	SCAN D-FLIP-FLOP
SDFFRHQX4/	SCAN D-FLIP-FLOP
SDFFRHQX8/	SCAN D-FLIP-FLOP
SDFFRX1/	SCAN D-FLIP-FLOP
SDFFRX2/	SCAN D-FLIP-FLOP
SDFFRX4/	SCAN D-FLIP-FLOP
SDFFRXL/	SCAN D-FLIP-FLOP
SDFFSHQX1/	SCAN D-FLIP-FLOP
SDFFSHQX2/	SCAN D-FLIP-FLOP
SDFFSHQX4/	SCAN D-FLIP-FLOP
SDFFSHQX8/	SCAN D-FLIP-FLOP
SDFFSRHQX1/	SCAN D-FLIP-FLOP
SDFFSRHQX2/	SCAN D-FLIP-FLOP
SDFFSRHQX4/	SCAN D-FLIP-FLOP
SDFFSRHQX8/	SCAN D-FLIP-FLOP
SDFFSRX1/	SCAN D-FLIP-FLOP
SDFFSRX2/	SCAN D-FLIP-FLOP
SDFFSRX4/	SCAN D-FLIP-FLOP
SDFFSRXL/	SCAN D-FLIP-FLOP
SDFFSX1/	SCAN D-FLIP-FLOP
SDFFSX2/	SCAN D-FLIP-FLOP
SDFFSX4/	SCAN D-FLIP-FLOP
SDFFSXL/	SCAN D-FLIP-FLOP
SDFFTRX1/	SCAN D-FLIP-FLOP
SDFFTRX2/	SCAN D-FLIP-FLOP
SDFFTRX4/	SCAN D-FLIP-FLOP
SDFFTRXL/	SCAN D-FLIP-FLOP
SDFFX1/	SCAN D-FLIP-FLOP
SDFFX2/	SCAN D-FLIP-FLOP
SDFFX4/	SCAN D-FLIP-FLOP
SDFFXL/	SCAN D-FLIP-FLOP
44	

Table 6 (continued) List of Standard Cells

List of Standard Cells	Description
EDFFHQX1/	ENABLE D-FLIP-FLOP
EDFFHQX2/	ENABLE D-FLIP-FLOP
EDFFHQX4/	ENABLE D-FLIP-FLOP
EDFFHQX8/	ENABLE D-FLIP-FLOP
EDFFTRX1/	ENABLE D-FLIP-FLOP
EDFFTRX2/	ENABLE D-FLIP-FLOP
EDFFTRX4/	ENABLE D-FLIP-FLOP
EDFFTRXL/	ENABLE D-FLIP-FLOP
EDFFX1/	ENABLE D-FLIP-FLOP
EDFFX2/	ENABLE D-FLIP-FLOP
EDFFX4/	ENABLE D-FLIP-FLOP
EDFFXL/	ENABLE D-FLIP-FLOP
12	
MDFFHQX1/	MUX D-FLIP-FLOP
MDFFHQX2/	MUX D-FLIP-FLOP
MDFFHQX4/	MUX D-FLIP-FLOP
MDFFHQX8/	
4	
SEDFFHQX1/	SCAN ENABLE D-FLIP-FLOP
SEDFFHQX2/	SCAN ENABLE D-FLIP-FLOP
SEDFFHQX4/	SCAN ENABLE D-FLIP-FLOP
SEDFFHQX8/	SCAN ENABLE D-FLIP-FLOP
SEDFFTRX1/	SCAN ENABLE D-FLIP-FLOP
SEDFFTRX2/	SCAN ENABLE D-FLIP-FLOP
SEDFFTRX4/	SCAN ENABLE D-FLIP-FLOP
SEDFFTRXL/	SCAN ENABLE D-FLIP-FLOP
SEDFFX1/	SCAN ENABLE D-FLIP-FLOP
SEDFFX2/	SCAN ENABLE D-FLIP-FLOP
SEDFFX4/	SCAN ENABLE D-FLIP-FLOP
SEDFFXL/	SCAN ENABLE D-FLIP-FLOP
12	
SMDFFHQX1/	SCAN MUX D-FLIP-FLOP
SMDFFHQX2/	SCAN MUX D-FLIP-FLOP
SMDFFHQX4/	SCAN MUX D-FLIP-FLOP
SMDFFHQX8/	SCAN MUX D-FLIP-FLOP
4	

List of Standard Cells	Description
SDFFHQX1/	SCAN D-FLIP-FLOP
SDFFHQX2/	SCAN D-FLIP-FLOP
SDFFHQX4/	SCAN D-FLIP-FLOP
SDFFHQX8/	SCAN D-FLIP-FLOP
SDFFNSRX1/	SCAN D-FLIP-FLOP
SDFFNSRX2/	SCAN D-FLIP-FLOP
SDFFNSRX4/	SCAN D-FLIP-FLOP
SDFFNSRXL/	SCAN D-FLIP-FLOP
SDFFQX1/	SCAN D-FLIP-FLOP
SDFFQX2/	SCAN D-FLIP-FLOP
SDFFQX4/	SCAN D-FLIP-FLOP
SDFFQXL/	SCAN D-FLIP-FLOP
SDFFRHQX1/	SCAN D-FLIP-FLOP
SDFFRHQX2/	SCAN D-FLIP-FLOP
SDFFRHQX4/	SCAN D-FLIP-FLOP
SDFFRHQX8/	SCAN D-FLIP-FLOP
SDFFRX1/	SCAN D-FLIP-FLOP
SDFFRX2/	SCAN D-FLIP-FLOP
SDFFRX4/	SCAN D-FLIP-FLOP
SDFFRXL/	SCAN D-FLIP-FLOP
SDFFSHQX1/	SCAN D-FLIP-FLOP
SDFFSHQX2/	SCAN D-FLIP-FLOP
SDFFSHQX4/	SCAN D-FLIP-FLOP
SDFFSHQX8/	SCAN D-FLIP-FLOP
SDFFSRHQX1/	SCAN D-FLIP-FLOP
SDFFSRHQX2/	SCAN D-FLIP-FLOP
SDFFSRHQX4/	SCAN D-FLIP-FLOP
SDFFSRHQX8/	SCAN D-FLIP-FLOP
SDFFSRX1/	SCAN D-FLIP-FLOP
SDFFSRX2/	SCAN D-FLIP-FLOP
SDFFSRX4/	SCAN D-FLIP-FLOP
SDFFSRXL/	SCAN D-FLIP-FLOP
SDFFSX1/	SCAN D-FLIP-FLOP
SDFFSX2/	SCAN D-FLIP-FLOP
SDFFSX4/	SCAN D-FLIP-FLOP
SDFFSXL/	SCAN D-FLIP-FLOP
SDFFTRX1/	SCAN D-FLIP-FLOP
SDFFTRX2/	SCAN D-FLIP-FLOP
SDFFTRX4/	SCAN D-FLIP-FLOP
SDFFTRXL/	SCAN D-FLIP-FLOP
SDFFX1/	SCAN D-FLIP-FLOP
SDFFX2/	SCAN D-FLIP-FLOP
SDFFX4/	SCAN D-FLIP-FLOP
SDFFXL/	SCAN D-FLIP-FLOP
44	

Table 6 (continued) List of Standard Cells

List of Standard Cells	Description
TLATNCAX12/	TRI-STATE LATCH
TLATNCAX16/	TRI-STATE LATCH
TLATNCAX2/	TRI-STATE LATCH
TLATNCAX20/	TRI-STATE LATCH
TLATNCAX3/	TRI-STATE LATCH
TLATNCAX4/	TRI-STATE LATCH
TLATNCAX6/	TRI-STATE LATCH
TLATNCAX8/	TRI-STATE LATCH
TLATNSRX1/	TRI-STATE LATCH
TLATNSRX2/	TRI-STATE LATCH
TLATNSRX4/	TRI-STATE LATCH
TLATNSRXL/	TRI-STATE LATCH
TLATNTSCAX12/	TRI-STATE LATCH
TLATNTSCAX16/	TRI-STATE LATCH
TLATNTSCAX2/	TRI-STATE LATCH
TLATNTSCAX20/	TRI-STATE LATCH
TLATNTSCAX3/	TRI-STATE LATCH
TLATNTSCAX4/	TRI-STATE LATCH
TLATNTSCAX6/	TRI-STATE LATCH
TLATNTSCAX8/	TRI-STATE LATCH
TLATNX1/	TRI-STATE LATCH
TLATNX2/	TRI-STATE LATCH
TLATNX4/	TRI-STATE LATCH
TLATNXL/	TRI-STATE LATCH
TLATSRX1/	TRI-STATE LATCH
TLATSRX2/	TRI-STATE LATCH
TLATSRX4/	TRI-STATE LATCH
TLATSRXL/	TRI-STATE LATCH
TLATX1/	TRI-STATE LATCH
TLATX2/	TRI-STATE LATCH
TLATX4/	TRI-STATE LATCH
TLATXL/	TRI-STATE LATCH
32	
DLY1X1/	DELAY
DLY1X4/	DELAY
DLY2X1/	DELAY
DLY2X4/	DELAY
DLY3X1/	DELAY
DLY3X4/	DELAY
DLY4X1/	DELAY
DLY4X4/	DELAY
8	
TIEHI/	TIE HIGH
TIELO/	TIE LOW
2	
FILL1/	FILLER CELL
FILL16/	FILLER CELL
FILL2/	FILLER CELL
FILL32/	FILLER CELL
FILL4/	FILLER CELL
FILL64/	FILLER CELL
FILL8/	FILLER CELL
7	

List of Standard Cells	Description
ANTENNA/	ANTENNA CELL
1	
HOLDX1/	MEMORY INPUT/OUTPUT LATCH
1	
VDD/	INHERITED CONNECTION POWER, “[VDD:%VDD!]”
1	
VSS/	INHERITED CONNECTION GROUND, “[VSS:%:VSS!]”
1	
ACHCONX2	ACCUMULATOR
1	

90nm CDK Access & Distribution

The 90nm CDK is undergoing continual development to expand it's capabilities as new flow requirements arise. As a result, not every aspect of the design kit has been tested completely with the various Virtuoso Flow Solutions flows.

Any user needing access to the 90nm CDK for internal development purposes can contact Kent Lewallen or Jeff Guy.

Due to the current development status, the 90nm CDK is not intended for distribution/release to Cadence customers or other 3rd parties at this time without the approriate NDA and license agreements. Any requests for 3rd party distribution should be provided to Jeff Guy.

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Frequently Asked Questions

QUESTION **What is the purpose of the Cadence CDK090?**
ANSWER *The Cadence 90nm complete design kit enables the 2H04 Ether and Raptor projects from which the AMS and Chip Integration custom IC flows, demos, & workshops will be created, tested, & delivered by internal Cadence groups.*

QUESTION **Who is the main contact person for the Cadence CDK090?**
ANSWER *For technical questions, contact Kent Lewallen. All other questions for the Cadence CDK090 can be directed to Jeff Guy.*

QUESTION **Does the Cadence CDK090 contain industry proprietary or non exportable data?**
ANSWER *No; however the CDK090 is the property of Cadence Design Systems, and can only be used with permission.*

QUESTION **What Cadence IP will be built using the CDK090 by Q404?**
ANSWER *The Ethernet Phy and Raptor wired communications controller (includes Condor MPU, Parrot 8 port Ethernet Phy, & Hummingbird PLL)*

QUESTION **Is a memory compiler part of the 90nm CDK?**
ANSWER *No, memory block and associated design tasks are considered to be part of the design IP associated with a particular reference design. We do not have a flow solution for designing memory blocks.*

QUESTION **Are there plans for multi-Vt cells for low leakage and low-power design?**
ANSWER *There are no plans at this time as these are not needed for the specific objectives.*

QUESTION **Are there plans to support FPGA design?**
ANSWER *There are no plans at this time as these are not needed for the specific objectives.*

QUESTION **Is there an AMS Siminfo section in the CDF (PDK) for primitives??**
ANSWER *Yes there is an AMS siminfo for each primitive in the cdf.* Also, if using ADE SPICE Netlist *Procedures (such as in SpectreDirect) these can be converted to AMS Netlist Procedures using the translator shipped with DFII.*

QUESTION **Are the simulation models BSIM3 or BSIM4?**
ANSWER BSIM3; however, our BSIM3 models leakage currents are similar to those of BSIM4

QUESTION **How would you compare the Cadence 90nm complete design kit with other industry kits?**

ANSWER *Though not completely tested, the process design kit is on par with industry standards; the standard cell library is only an implementation of the basic cells which meet minimum needs; similarly, the I/O library contains only a few basic cells necessary for minimum needs.*

QUESTION **Does the Cadence CDK090 support the proper use of inherited connections?**

ANSWER *Yes; the PDK, standard cells, and I/O cells in the CDK fully support the Cadence-recommended inherited connections implementation. The provided cells have power and ground "net expressions" of "[VDD:%:VDD!]" and "[VSS:%:VSS!]" respectively.*